Junbo Zhou

Ph.D. Student in Electrical Engineering · Southwest Jiaotong University

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Google Scholar | ORCID: 0009-0007-5154-1893 | ResearchGate

Research Interests

Hyperspectral image processing; deep learning; spectral-spatial representation learning; classification, unmixing, and super-resolution; efficient neural networks; hyperspectral insulator condition assessment.

Education

Southwest Jiaotong University | September 2023 – Present

Ph.D. in Electrical Engineering (in progress), Chengdu, China.
Supervisor: Prof. Guoqiang Gao.

Wuhan Polytechnic University | September 2020 – June 2023

Master’s degree in Electronic and Information Engineering, Wuhan, China.
Supervisor: Prof. Shan Zeng. GPA: 87.06/100.

Tiangong University | September 2016 – June 2020

Bachelor’s degree in Information and Computational Science, Tianjin, China.
GPA: 85.68/100.

Research Experience

Outdoor Hyperspectral Characterization and Recognition of Insulator Contamination

September 2023 – Present | Doctoral research

Design outdoor hyperspectral experiments and datasets under varying illumination, weather, and observation-distance conditions. Develop and evaluate methods for spectral-spatial representation learning, contamination-level recognition, hyperspectral unmixing, and lightweight inference.

Hyperspectral Image Classification with 3D Convolutional Neural Networks

September 2020 – June 2023 | Master's research

Designed and implemented the classification models and conducted ablation, parameter, and comparative experiments on multiple benchmark datasets. Prepared the resulting journal publications.

Interaction Characteristics and Analysis Methods for Electricity, Hydrogen, and Thermal Multi-Energy Flows

June 2025 – Present | Project core member

National Key Research and Development Program of China | Grant No. 2024YFB2606300
Project leader: Prof. Guangning Wu

Contributed to proposal preparation and technical roadmap design. Ongoing work involves developing coupled models and data-driven methods to characterize energy transmission, conversion, and interactions across multiple timescales.

Journal Publications

1. A Hyperspectral Method for Joint Assessment of Aging and Contamination States in Composite Insulators

H. Li, Y. Ou, Y. Guo, Y. Fan, J. Zhou, et al.
IEEE Transactions on Dielectrics and Electrical Insulation, 2026.
DOI: 10.1109/TDEI.2026.3712704

2. Fast contamination recognition network: a knowledge distillation based hyperspectral outdoor insulator contamination level recognition network

J. Zhou, G. Gao, Y. Guo, et al.
Expert Systems with Applications, 2026, vol. 300, article 130401.
DOI: 10.1016/j.eswa.2025.130401

3. A pixel-level assessment method of the aging status of silicone rubber insulators based on hyperspectral imaging technology and IPCA-SVM model

Y. Fan, Y. Guo, Y. Liu, S. Xiao, J. Zhou, et al.
Expert Systems with Applications, 2025, vol. 263, article 125788.
DOI: 10.1016/j.eswa.2024.125788

4. Optical imaging technology application in transmission line insulator monitoring: A review

Y. Liu, Y. Guo, Y. Fan, J. Zhou, et al.
IEEE Transactions on Dielectrics and Electrical Insulation, 2024, vol. 31, no. 6, pp. 3120–3132.
DOI: 10.1109/TDEI.2024.3351093

5. A Novel Spatial–Spectral Pyramid Network for Hyperspectral Image Classification

J. Zhou, S. Zeng, G. Gao, et al.
IEEE Transactions on Geoscience and Remote Sensing, 2023, vol. 61, pp. 1–14, article 5519314.
DOI: 10.1109/TGRS.2023.3303338

6. An enhanced spectral fusion 3D CNN model for hyperspectral image classification

J. Zhou, S. Zeng, Z. Xiao, et al.
Remote Sensing, 2022, vol. 14, no. 21, article 5334.
DOI: 10.3390/rs14215334

Accepted Conference Paper

HSRNet: Hyperspectral Retentive Network for Hyperspectral Image Classification

J. Zhou, G. Gao, C. Zhang, et al.
IEEE International Joint Conference on Neural Networks (IJCNN), 2026. Accepted.

Manuscripts Under Review

1. Recurrent Spectral State Propagation for Hyperspectral Image Super-Resolution

J. Zhou, G. Gao, S. Zeng, et al. Manuscript under review.

2. ERRNet: Recurrent Endmember Representation Learning for Hyperspectral Unmixing

J. Zhou, G. Gao, Y. Guo, et al. Manuscript under review.

3. Amplitude–Shape Analysis of Insulator Contamination Spectra across Sunny and Cloudy Conditions

J. Zhou, G. Gao, Y. Guo, et al. Manuscript under review.

4. Observation-Distance Effects on Ground-Based Hyperspectral Assessment of Insulator Contamination: Spatial–Spectral Degradation and Effective Distance Estimation

J. Zhou, G. Gao, Y. Guo, et al. Manuscript under review.

5. SpecRetNet: A Novel Spectral-Spatial Retention Network for Low-Irradiance and Extended-Standoff Hyperspectral Insulator Contamination Recognition

J. Zhou, G. Gao, Y. Guo, et al. Manuscript under review.

Academic Service

Peer reviewer for IEEE Transactions on Geoscience and Remote Sensing, Engineering Applications of Artificial Intelligence, Pattern Recognition, Scientific Data, and other international journals.

Honors and Awards

2023 — Yanghua Emerging Talent Scholarship, Southwest Jiaotong University.

2023 — Outstanding Master's Thesis Award, Wuhan Polytechnic University.

2023 — Outstanding Graduate Student, Wuhan Polytechnic University.

2022 — Third Prize, 19th China Post-Graduate Mathematical Contest in Modeling (Huawei Cup).

2022 — Outstanding Graduate Student Award, Wuhan Association for Computer Software Engineering.

2021–2022 — Yinianyuan Scholarship and Outstanding Graduate Student Role Model, Wuhan Polytechnic University.

2021 — Third Prize, 18th China Post-Graduate Mathematical Contest in Modeling (Huawei Cup).

2021 — Second Prize, 15th Graduate Zhixing Forum, Wuhan Textile University.

Skills and Languages

Programming and tools: Python, Linux, Microsoft Office, Adobe Illustrator, and Adobe Photoshop.

Languages: Mandarin (native); English (CET-6).